Exploiting Rigid-Body Symmetry in Flow-Matching Motion Planning

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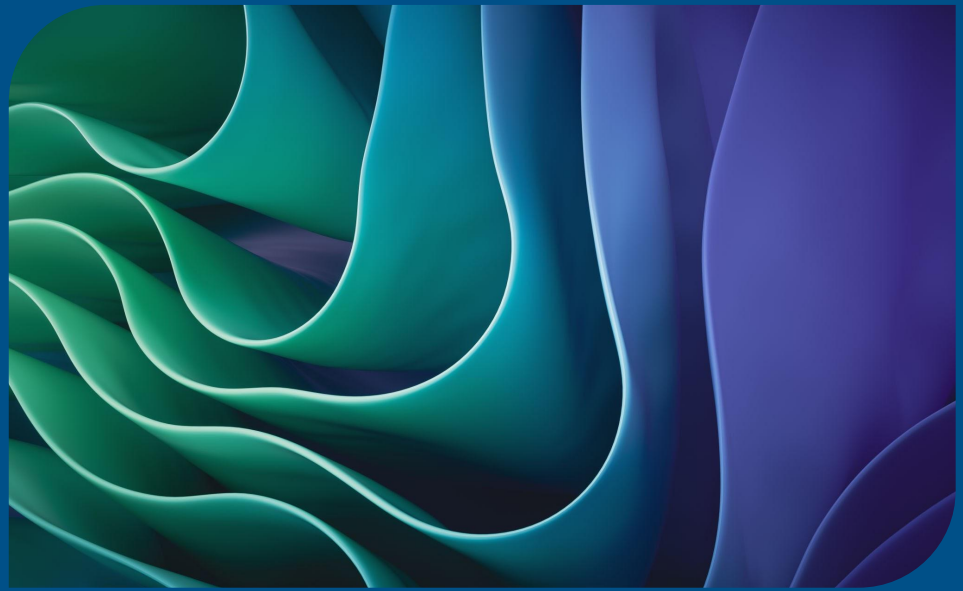


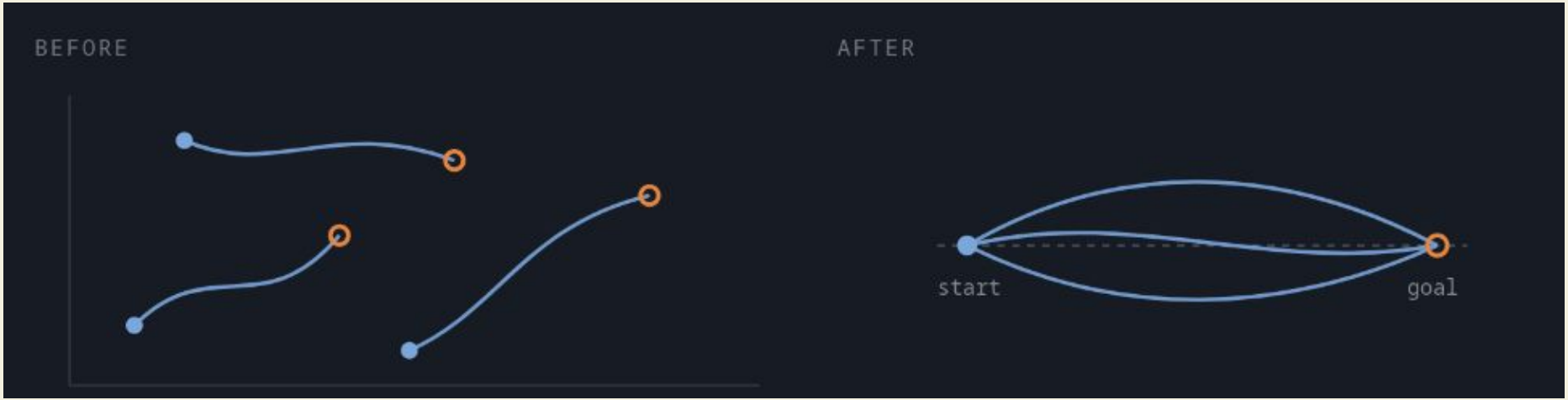
1. 3D Benchmark
2. Baseline
3. Explore Canonicalization
4. Theoretical Validation
5. Discussions & Conclusions

Flow Matching

Flow Matching is a generative modeling algorithm, that has a model simulate a vector field, such that we give as input some noisy data probability distribution, and as output we get the probability distribution we desire.

Why a distribution? Instead of predicting a single deterministic path, it is better if we learn a conditional generative policy.

We model trajectory generation as a continuous-time flow


BEFORE
AFTER
start
goal

But in 3 dimensions..

There's one spin you can't get rid of

You can line up start and goal. But the room can still be rolled around that line like meat on a rotisserie, and there is no consistent way to pick which way is "up".

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3D Benchmarj

	collision-free	s/query
straight line	15.6%	—
RRT-Connect	99.6%	0.284
expert (RRT+CHOMP)	100%	0.342
learned planner	*see next slides*	**0.067**

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Hello~


15% → 51%, from a coordinate change
Before
14.6
Straight line, eyes shut
15.6
After
51.1

Built an Expert Planner

RRT-Connect, random shortcutting, uniform resampling, and CHOMP refinement

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Equivariance can be put into the data (roll augmentation), the operator (frame averaging at inference), or the weights (equivariant architecture).

mechanism	gain
roll augmentation	+0.8
frame averaging (K=1→3)	+0.68 ± 0.13
equivariant architecture	*next slide*

- Instead of *hoping* it learns the symmetry, I rebuilt it so spinning the room **automatically** spins the answer
- It's a guarantee, not a hope — true even before any training at all
- Same final score. But it gets there **4× faster**

AFTER…	NEW NET	OLD NET
10 rounds	**35.8**	**22.2**
40 rounds	44.6	37.9
300 rounds	50.9	50.8

On a whim I told the model, at every point along the path, **how close the nearest obstacle is and which way it is**. It already had that information in its input — it just wasn't using it.

	WITHOUT	WITH
Before the trick	14.6	running…
After the trick	**50.8**	**79.2**
+ the new net	**50.9**	**82.2**

+28 points. Bigger than everything else I did this summer, combined.

And the fancy network, worth **+0.1** on its own, became worth **+3** once it had something to work with.

Still open. One box is still training. If it comes back high, part of my earlier story needs rewriting — and I'd rather find that out myself.

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:)

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Drawing a Conclusion

The best idea of the summer took one afternoon and small compute power. The thing I spent six weeks on was worth a tenth of a point